HOMEWORK NR. 2

Section Centroid and Moments of Inertia

Student code: 243483

Given

Last digit $A = 3 \Rightarrow$ Shape 3
Last but one digit $B = 8$

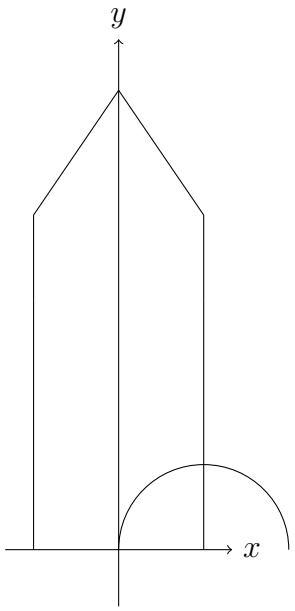
$$a = 11 \text{ cm}, \qquad b = 15 \text{ cm}$$

$$R = \frac{b}{2} = 7.5 \text{ cm}$$

1. Geometry of the Section

The section consists of:

- Semicircle (radius R)
- Rectangle (width b, height $2a$)
- Triangle (base b, height a)



2. Areas

Semicircle

$$A_1 = \frac{1}{2}\pi R^2 = \frac{1}{2}\pi(7.5)^2 = 88.36 \text{ cm}^2$$

Centroid location from bottom:

$$y_1 = R - \frac{4R}{3\pi} = 7.5 - 3.183 = 4.317 \text{ cm}$$

Rectangle

$$A_2 = b(2a) = 15 \cdot 22 = 330 \text{ cm}^2$$

$$y_2 = R + a = 7.5 + 11 = 18.5 \text{ cm}$$

Triangle

$$A_3 = \frac{1}{2}ba = 82.5 \text{ cm}^2$$

$$y_3 = 29.5 + \frac{a}{3} = 33.167 \text{ cm}$$

Total Area

$$A = 88.36 + 330 + 82.5 = 500.86 \text{ cm}^2$$

3. Centroid

$$S_x = \sum A_i y_i$$

$$S_x = 88.36(4.317) + 330(18.5) + 82.5(33.167)$$

$$S_x = 9222.7$$

$$\bar{y} = \frac{S_x}{A} = \frac{9222.7}{500.86} = 18.42 \text{ cm}$$

Because of symmetry:

$$\bar{x} = 0$$

$$\boxed{(\bar{x}, \bar{y}) = (0,\ 18.42 \text{ cm})}$$

4. Moment of Inertia about x-axis

$$I_x = \sum \left(I_{x,c} + Ad^2\right)$$

Semicircle

$$I_{x,c1} = \frac{\pi R^4}{8} - \frac{8R^4}{9\pi} = 346.97$$

$$d_1 = 14.103$$

$$I_{x1} = 17917$$

Rectangle

$$I_{x,c2} = \frac{bh^3}{12} = 13310$$

$$I_{x2} = 13312$$

Triangle

$$I_{x,c3} = \frac{ba^3}{36} = 554.6$$

$$I_{x3} = 18480$$

Total

$$I_x = 17917 + 13312 + 18480$$

$$\boxed{I_x = 4.97 \times 10^4 \text{ cm}^4}$$

5. Moment of Inertia about y-axis

Semicircle

$$I_{y1} = \frac{\pi R^4}{8} = 1242.77$$

Rectangle

$$I_{y2} = \frac{hb^3}{12} = 6187.5$$

Triangle

$$I_{y3} = \frac{b^3 a}{48} = 773.4$$

$$\boxed{I_y = 8.20 \times 10^3 \text{ cm}^4}$$

6. Principal Moments

Since the section is symmetric about the vertical axis:

$$I_1 = I_x$$

$$I_2 = I_y$$

$$\boxed{I_1 = 4.97 \times 10^4 \text{ cm}^4}$$

$$\boxed{I_2 = 8.20 \times 10^3 \text{ cm}^4}$$